Closing the inert consecutive seed

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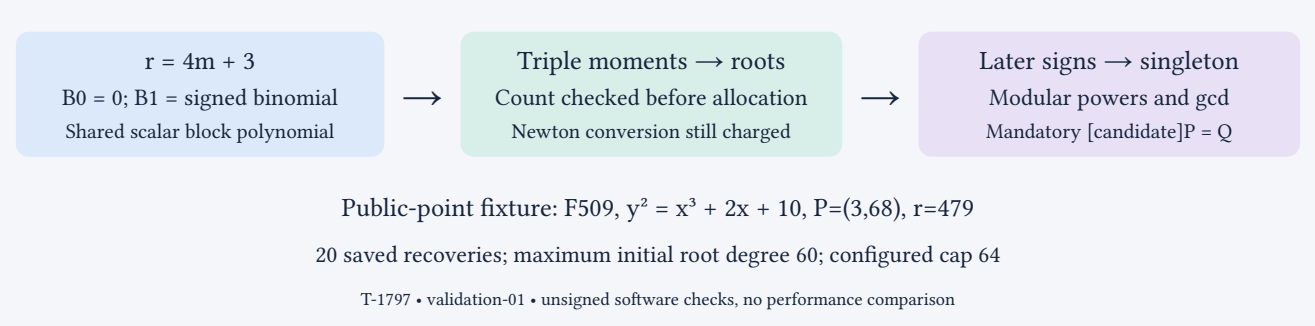
The missing weighted seed for consecutive signs at scalar primes r=3 modulo4 now reduces to a central binomial coefficient. A native scalar polynomial-block evaluator supplies it, and a new adapter connects triple moments to public-point recovery under explicit root-degree and query caps.



Derivation and implementation

Write r=4m+3 and h=2m+1. Substitution z=X+1 gives the cubic z³−z. Its unweighted character sum vanishes by odd symmetry. In the weighted sum, finite-field power sums select the coefficient with index m, giving B1=(−1)^(m+1) binom(2m+1,m), with B0=0.

Evaluate the binomial as the product of m+2 through2m+1 divided by m factorial. One polynomial P(Y)=product of (Y+i), i=1..w, w=ceil(sqrt(m)), serves both block progressions. Existing native product/remainder trees evaluate them together. Only the shorter tails are multiplied individually; one checked inversion completes the ratio. Primality is caller-certified. Width and evaluation-count caps precede tree allocation.

The adapter retains the earlier CM branch for r=1 modulo4. It handles zero signs at each of the first three offsets through forward-verified endpoints, initializes roots from exact triple moments, and applies subsequent modular-power/gcd constraints. Every returned scalar passes its point equality check.

Native checks

Three selected methods pass; 160 imported methods are filtered. Nineteen primes through100003 match separate linear-product seed references. Eighty-eight sign classes across eleven primes match846 moments and exact root polynomials against enumerated references. At least one seed fixture exercises Karatsuba and Newton code paths.

All20 saved public-point targets on the displayed curve recover, with maximum initial root degree60 under cap64. A separate target −2P recovers477 through the third zero sign. Wrong candidates, two-query and degree16 limits, invalid coordinates and seed allocation caps have controls. Test/normal builds retain17/24 warning lines, including summaries. There were no compilation or test failures. The original4,563-case cohort is unchanged.

Cost and remaining construction

Seed setup is target-independent and reusable, but remains charged. With multiplication cost M, the conservative seed bound is O(M(sqrt(r)) log r + log r) field work and O(sqrt(r) log r) words. Installed Karatsuba implies exponent $\log_2(3)/2$; no faster FFT bound or measured speedup is attributed to this code.

Root initialization still requests all candidate moments and uses O(m_candidates²) Newton work, with candidate count generally of order r/8. Source evaluation and later gcd costs remain. Next extend the moment representation through additional signs without materializing that root set, and investigate more direct seed computation.

Sources and custody

Bostan–Gaudry–Schost (2007), recurrence and Cartier–Manin setup, informs the established block arithmetic. Existing blocked-character, pair/triple and polynomial modules were inspected; KB unavailable. DERIVATION.md contains proof and source links. The28 source/input pins, commands, logs, extraction hashes, manifest and this inspected report are included. No independent signed verdict, timing panel or cloud job.